

Deployio

AI-Powered Platform for Automated Application Deployment & DevOps Management

Project Type: Final Year Project **Domain:** Cloud Computing & Software Engineering **Platform:** deployio.tech

ABSTRACT

Deployio is a web-based deployment platform that helps developers publish applications to the internet without deep expertise in servers or manual DevOps workflows. Users connect their code repository (GitHub, GitLab, or similar), follow a guided seven-step wizard, and let the system automatically analyze the codebase, suggest configuration, and register the project. From a central dashboard, users deploy new versions, track live build progress, view deployment history, and manage releases. The platform unifies repository integration, intelligent codebase understanding, automated hosting, and lifecycle management—making industry-standard release practices accessible to students, startups, and independent developers.

OBJECTIVES

- To design and implement an end-to-end system from project onboarding to live deployment and post-release management.
- To simplify deployment through a guided wizard that replaces fragmented manual setup steps.
- To automatically analyze repositories and recommend suitable build and runtime settings using AI-assisted stack detection.
- To integrate Git providers securely and automate packaging, build, and release with real-time status updates.
- To provide a unified dashboard for projects, deployments, analytics, and operational actions.
- To demonstrate a scalable microservices architecture suitable for production-style workloads.

END-TO-END FLOW



TOOLS & TECHNOLOGY

Frontend

React 19, Vite, Redux Toolkit, Tailwind CSS, React Router, Framer Motion, Socket.IO Client, Recharts

Backend API

Node.js, Express.js, MongoDB (Mongoose), Redis, JWT & OAuth2 (GitHub, GitLab, Google), Passport.js, Socket.IO

AI & Analysis Service

Python, FastAPI, Uvicorn, Pydantic, Groq / OpenAI / Gemini APIs, Redis caching

Deployment Agent

Python, FastAPI, Docker, GitPython, MongoDB Motor driver, real-time WebSockets

Infrastructure & DevOps

Docker, Traefik, Nginx, VPS hosting, GitHub Actions / CI pipelines, environment-based configuration

Supporting Tools

Git, VS Code, Swagger API docs, Winston logging, Cloudinary, Nodemailer

RESULTS & OUTCOMES

- Delivered a production-oriented platform with separate frontend, API server, AI analysis service, and deployment agent.
- Implemented a complete 7-step project creation wizard with automated analysis and pre-filled settings.
- Enabled deployments with live progress tracking and centralized deployment history.
- Built dashboard modules for projects, deployments, analytics, integrations, and monitoring.
- Shipped sample demo applications and deployed the platform live at deployio.tech.

7 Wizard Steps	4+ Microservices	15+ Languages Supported	~30s Avg. Deploy Time
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